Automotive Cyber Security Academy 2023

Basic Information Technology Security

Topic 4: Security Threats

Automotive Cyber Security Academy

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Outline

- Security Threats
- Defensive Measures

Note:

- Please recap the topic 2

Security Threats

Security Threats

- Malware:
  - Several classes
- Denial of Service (DoS)
- Man-in-The-Middle (MiTM)
- Phishing
- Pharming
- IP Spoofing
- Eavesdropping
- Spam
- Social Engineering

Security Threats - Malware

- Includes all the malicious software, which aim to obtain credentials, to cause impact, to be employed as means for other threats.

- The associated attacks can include:
  - Exploitation of well-know vulnerabilities
  - Zero-day attacks, which exploit vulnerabilities that are not known.

Security Threats – Malware Classes

- **Trojan** – Appears to be benign software but, after its execution by the user opens a backdoor for access by hacker(s).
- **Virus** – Malware that injects malicious code into legitimate programs.
- **Worm** – Can replicate by itself, without the need to infect a host program or file, often spreads without any user action.
- **Bot / Botnet** – Uses multiple infected computers under the command of an attacker. The infected computers connect to command and control (C&C) servers. Use to trigger other type of attacks

Security Threats – Malware Classes

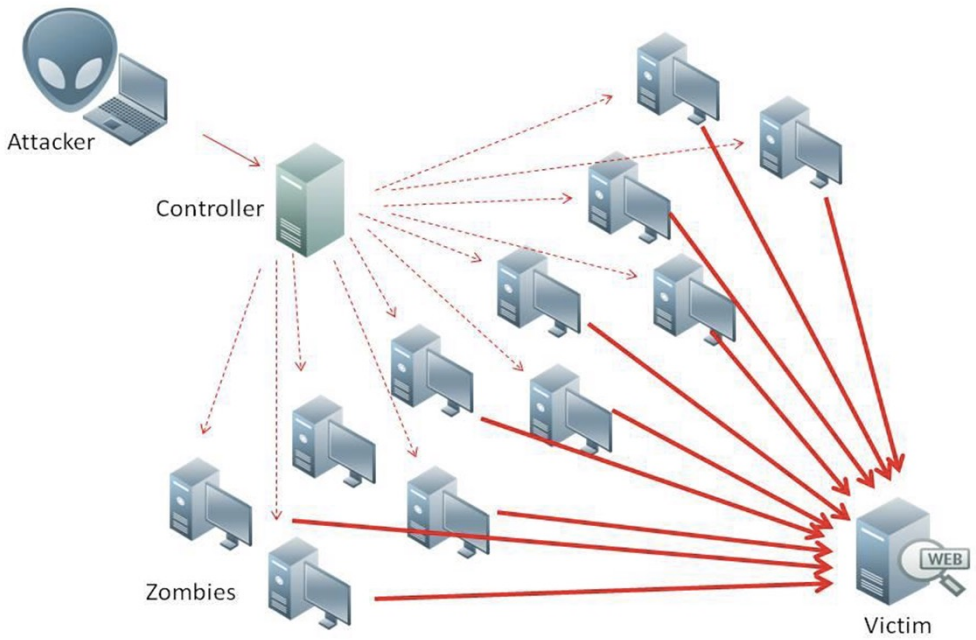
- **Adware** – Software that displays Advertisements in the user's web browser when it is not supposed to happen.
- **Keylogger** – Software to record all the keystrokes.
- **Ransomware** – Infects the computed and encrypts the stored data and demands for a payment ('ransom') to decrypt it.
- **Spyware** – Software that monitors user activity and that reports this activity to the attacker

Security Threats – Malware Classes

- **Rootkit** – Used in Unix-based systems to gain root privileges.
- **Remote Access Tool (RAT)** – software which purpose is to provide remote access to a computer. It may be used by botnets or others.

Security Threats – Distributed Denial of Service (DDoS)

- **Denial of Service (DoS)** – Aims to make a machine/service unavailable, and this is done by overwhelming system resources.



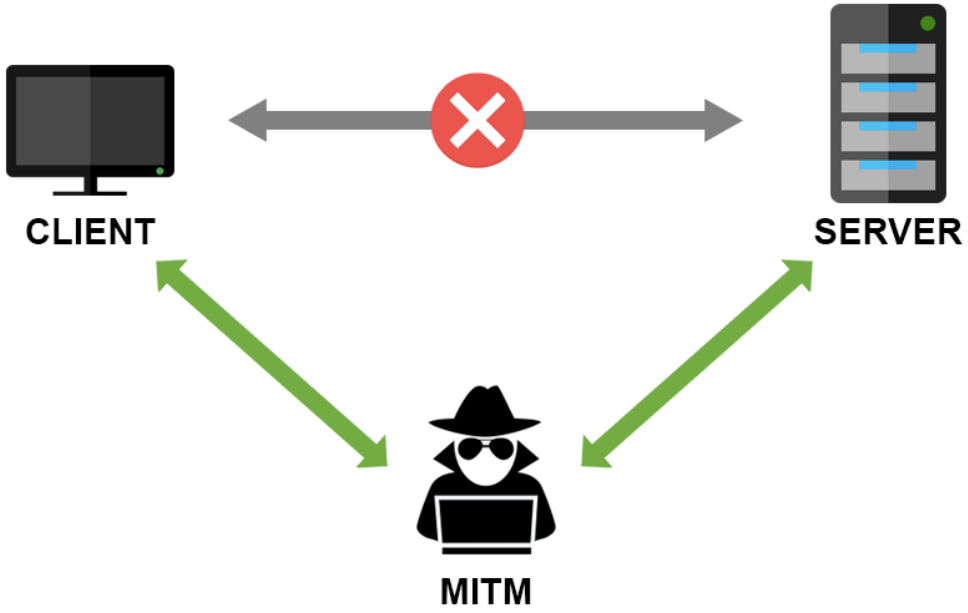
Source: Wikipedia

Security Threats – Distributed Denial of Service (DDoS)

Source: [Checkpoint]

Security Threats – Man-in-The-Middle (MiTM)

- **Man-in-The-Middle (MiTM)** – Aims to hijack a session between a trusted client and a network server, to intercept the communication between them.



- Example: ARP spoofing

Security Threats – Phishing

- **Phishing** – Attack where it is send an email that appears to be from trusted sources with the goal of gaining personal information or influencing users to perform an action.

- Example: fake online banking messages to make the user to install malware

Security Threats – Pharming

- **Pharming** – Phishing Attack where the user is redirected to an fraudulent URL. This attack can be performed by the modification of DNS data.

- Example: Instead of accessing google.com, would access adventureinsecurity.com.

Security Threats – IP Spoofing

- **IP Spoofing** – Spoofing the information of IP addresses used in IP packets with illegitimate sources. This leads that the replies are sent not to the legitimate source but to the attacker.

- Example:

Security Threats – Eavesdropping

- **Eavesdropping** – interception of network traffic to obtain passwords, confidential information. The types of attacks can be:
  - Passive : the message transmission in the network is listened by an attacker
    - **Sniffing** – Using specific programs to listen illegally the data that is being exchanged between two entities that are communicating.
  - Active: the Attacker actively grabs the information by distinguishing himself as a friendly unit and by sending queries to transmitters, kind of probing, scanning.

Security Threats – Social Engineering

- **Social Engineering** – Malicious activities accomplished through human interactions. Uses psychological manipulation to trick users into making security mistakes or giving away sensitive information.
- Happens in several steps.

Defensive Measures

Defensive Measures – What we can do

- Distrust and doubt
- Access control and non-repudiation:
  - Authentication, Authorization, Accounting (AAA) mechanisms
- Data Protection:
  - Encryption
  - Digital Signatures

Defensive Measures

- Authentication:
  - Strong Passwords changed frequently
  - Avoid reusing passwords
  - Use password managers (Bitwarden)
  - Use multi factor authentication

Defensive Measures

- Infrastructure/device protection:
  - Vulnerability Scanners
  - Intrusion Detection System (IDS)
  - Endpoint protection (Anti-virus, Host IDS)
  - Network Protection (Firewalls, Network IDS)
  - Honeypots
  - Intrusion Prevention Systems

Reflection

Reflections

- Do you know any victim of such threats ?
- How they have proceeded/recovered ?

Thanks

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The NEW and IMPROVED - ACSA 2023 is here!